

- **Object recognition:** These include highly advanced computer vision, NLP, or algorithms that, in machine learning, help to recognise or categorise objects, entities, or patterns. For instance, in a warehouse, a robot can distinguish between boxes, employees, and vehicles while it moves to ensure safe passage.

Cognitive module: The cognitive module serves as the 'brain' of the agentic AI system. It creates goals and constructs the plan for execution by making decisions.

Goal representation: In this module, an AI agent defines the goals for demonstration. For example, the goal for an AI assistant like Alexa is to control the thermostat, play music, or ask users a question. Goals also include long-term planning, such as optimising energy use in a smart grid system.

- **Planning:** Action plans and strategies are formulated from the current state of the environment. For instance, a delivery drone determines its optimal path with parameters like weather, air traffic, and battery limitations.
- **Decision-making:** The module selects among alternatives, choosing the best one. An example would be a computer program considering patient symptoms and deciding that tests or some form of treatment are required. Decisions most often involve a trade-off among incompatible goals.

Action module: The action module makes decisions and thus enables an AI agent to perform an action effectively.

- **Actuators:** These allow an agentic AI system to execute in a physical or virtual way. Motors in robotics and the use of APIs or network interfaces for the transfer of instructions in virtual systems are some examples.
- **Execution:** This is the sub-component that implements the chosen action. For instance, a

vacuum robot charts a cleaning path, while a customer support chatbot responds to the user's query with a customised message.

User module: This module facilitates humans to interact with the AI system and ensures that the agent is compatible with user demands and expectations.

The AI system harvests the preferences or commands from users in different interfaces, including voice commands, touchscreens, or keyboards. An example is the recommendation system in Netflix which keeps learning about the users' viewing habits based on their ratings and changes its recommendations based on that.

AI agents module: In complex systems, many AI agents collaborate to achieve common objectives. This module controls the interactions and coordination among such agents.

Sharing information among AI agents leads to better overall

performance. For example, traffic management and public transport AI systems for smart cities work together to optimise travel times and reduce traffic within the city.

Multi-agent systems support decision-making by multiple agents, which improves efficiency and scalability. Dedicated agents can focus on conducting unique duties. One AI system may be set for e-commerce customer service while another handles inventory forecasting.

How does agentic AI work?

Perception: The perception module gathers inputs from the environment via sensors. These are processed to yield meaningful images, sounds, or other numeric values from which an agent can recognise and understand the world. A virtual assistant, for example, takes speech inputs or text data while a robot utilises vision sensors to identify any possible obstacles.

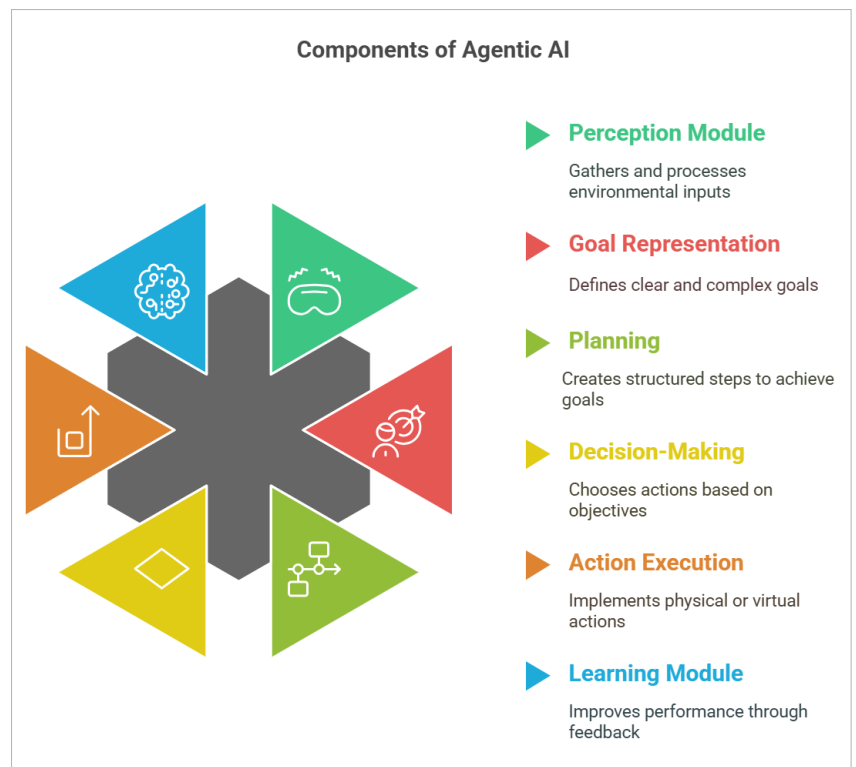


Figure 4: Components of agentic AI